Closing the Gate

Nikita Matveev

Algorithm

A* chooses the next cell using g + h. Here, g is the path length so far and h is Manhattan distance to the goal. The search also remembers whether Mike has found Eleven and collected the Blindfold. Before finding Eleven, the estimate includes the distance to her and then to the Gate.

The program first explores known safe cells. It leaves the Blindfold for later so it can finish checking the area without protection. If it must pick up the Blindfold to reach Eleven, it waits before entering her cell instead. Protection starts only after both have been found.

After exploring, A* finds the shortest mission from the start through Eleven to the Gate. Mike then walks to the Gate and prints this length. Exploration moves are counted separately. If the mission is impossible, the answer is -1. Variant A has view radius 1 and variant B has radius 2.

PEAS

Performance: reach the Gate with Eleven, use a shortest safe route and avoid death.
Environment: an 11 x 11 grid with enemies, items and hidden cells.
Actuators: move up, down, left or right; send the final answer.
Sensors: nearby objects and danger cells, Eleven's starting position and the Gate message.

Testing

Both variants were tested on the same 1000 random maps, using seed 2026. Maps have 1-2 Demogorgons, 1-2 Army squads and one Vecna. The start and item cells are safe. Invalid placements are rejected, but valid maps with no route are kept.

Time includes the solver and local observations, but not map generation or the BFS answer check. These are local Python 3 timings. A win means a possible mission; a loss means no solution. A correct -1 is not a program failure.

Result	A (radius 1)	B (radius 2)
Mean time, ms	4.340	5.487
Median time, ms	4.045	5.156
Mode time, ms	0.007, 2.918	0.010
Time standard deviation, ms	1.972	2.464
Wins / losses	926 / 74	926 / 74
Win / loss percentage	92.6% / 7.4%	92.6% / 7.4%
Win: mean / median / mode / SD	0.926 / 1 / 1 / 0.262	0.926 / 1 / 1 / 0.262
Loss: mean / median / mode / SD	0.074 / 0 / 0 / 0.262	0.074 / 0 / 0 / 0.262

Time modes use rounding to 0.001 ms. For win/loss statistics, each map gets a 0 or 1. SD is the population standard deviation.

Comparison

Both variants solved 926 out of 1000 maps (92.6%). They returned -1 on the same 74 maps (7.4%). This makes sense because the view radius changes what Mike can see, but it does not change the enemies or the safe routes on the map.

The Testing table shows that A was faster on average: 4.340 ms against 5.487 ms for B. The median was also lower for A (4.045 ms against 5.156 ms). B had a larger standard deviation (2.464 ms against 1.972 ms), so its run times varied more.

A larger view does not always make this program faster. The program still explores the reachable area, while each observation in B contains more cells to check. This may explain why B took more time in this local test. All 2000 answers were checked against BFS and matched it.

Unsolvable maps

There were 74 unsolvable maps out of 1000 (7.4%). Both variants had the same cases and correctly returned -1. On 71 maps, Mike could not reach Eleven. On the remaining 3, he could reach Eleven but could not finish the mission at the Gate.

The two maps below are typical examples of these two failure cases. Only two examples are shown here; there were 74 unsolvable maps in the full test.

Map 5

	0	1	2	3	4	5	6	7	8	9	10
0	M	#	.	.	.	.	.	.	.	.	.
1	#	A	#	.	.	.	.	.	.	.	.
2	.	#	.	.	.	.	.	.	.	.	.
3	.	.	.	.	.	.	.	.	#	.	.
4	#	#	.	.	#	#	#	#	A	#	.
5	D	#	.	.	#	D	#	.	#	.	.
6	#	#	.	.	#	#	#	.	.	.	.
7	.	.	.	.	.	.	.	.	.	.	.
8	.	G	.	#	#	#	#	#	.	.	.
9	.	.	B	#	#	#	#	#	.	.	.
10	E	.	.	#	#	V	#	#	.	.	.

Map 63

	0	1	2	3	4	5	6	7	8	9	10
0	M	.	.	.	.	.	.	.	.	.	.
1	#	#	#	.	#	.	.	.	.	.	.
2	#	D	#	#	A	#	.	#	#	#	.
3	#	#	#	#	#	#	#	#	D	#	.
4	.	.	.	#	#	#	#	#	#	#	.
5	.	.	.	#	#	V	#	#	.	E	.
6	.	.	.	#	#	#	#	#	.	.	.
7	.	.	.	#	#	#	#	#	.	.	.
8	.	.	B	.	.	#	.	.	.	.	.
9	G	.	.	.	#	A	#	.	.	.	.
10	.	.	.	.	.	#	.	.	.	.	.

M: start, E: Eleven, G: Gate, B: Blindfold. D/A/V: enemies. # means a dangerous cell; a dot means a safe cell before protection.

Map 5: the Army squad at (1, 1) makes both (0, 1) and (1, 0) dangerous. Mike starts at (0, 0), so he cannot make even the first move and cannot reach Eleven.

Map 63: Mike can reach Eleven, but the Blindfold and Gate are outside the area he can reach safely. Since he cannot collect the Blindfold, protection never starts and there is no safe way to the Gate.

Overlapping objects, coordinates outside the field and an unsafe start or item cell are invalid placements. They were not included in the 1000-map test.